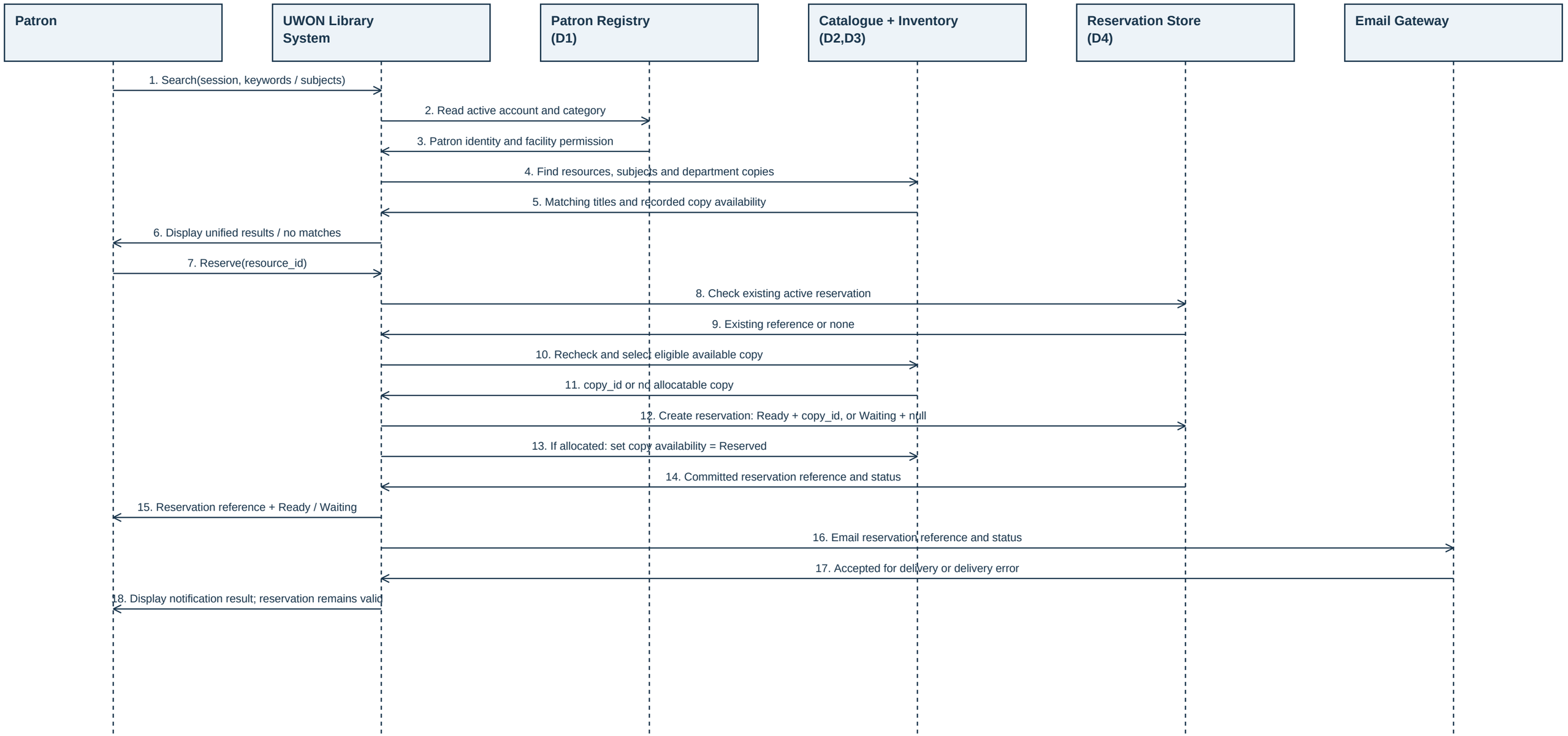
# S1 — Event Trace Diagram

Search the unified catalogue and reserve a resource

An active, authorized patron searches across UWON department holdings using the shared subject vocabulary, reviews recorded availability, and reserves a resource. If an eligible copy is available, the same transaction allocates it and marks the reservation Ready; otherwise the reservation is Waiting without an allocated copy. The patron receives a reference and an email notification attempt, while duplicate active reservations return the existing reference.




### Scenario conditions and alternatives

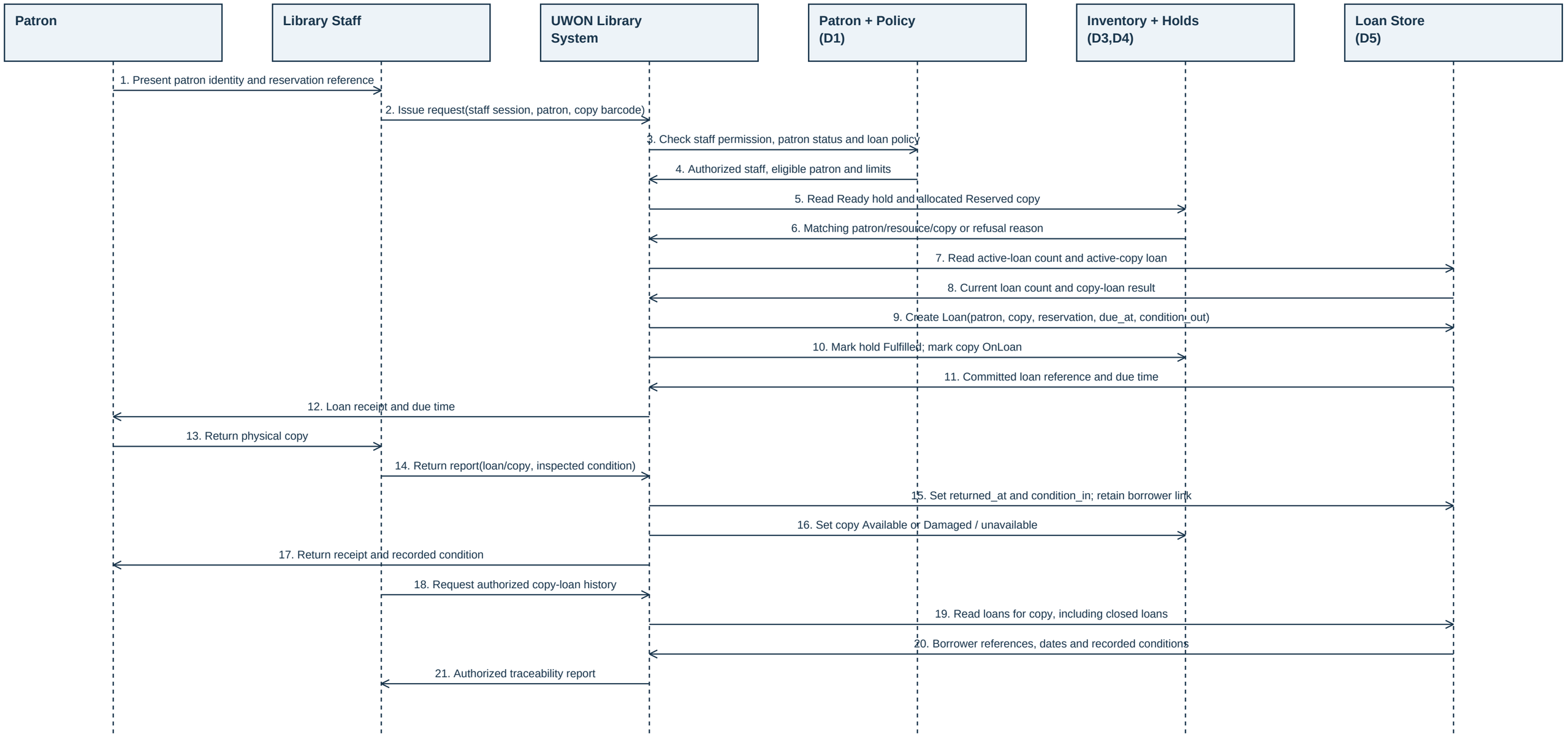
Precondition: registered active patron and facility permission. Reservation creation and any copy allocation are one atomic transaction; copies marked Binding, Missing, Damaged or OnLoan cannot be allocated. Alternatives: invalid permission -> deny without writes; no search match -> empty results; no available copy -> Waiting; duplicate hold -> existing reference; email failure does not cancel a committed reservation.

Time proceeds downward. Logical repositories refer to the same persistent data groups used in the ER and DFD views.

# S2 — Event Trace Diagram

Borrow a reserved copy, return it and preserve borrower history

A patron presents a Ready reservation, and authorized library staff scan its allocated copy. The system checks account status, the applicable loan policy and reservation ownership before recording the loan and due time. On return, staff record the copy condition, the loan is closed without deleting borrower history, and the copy becomes Available or is withheld as Damaged.




### Scenario conditions and alternatives

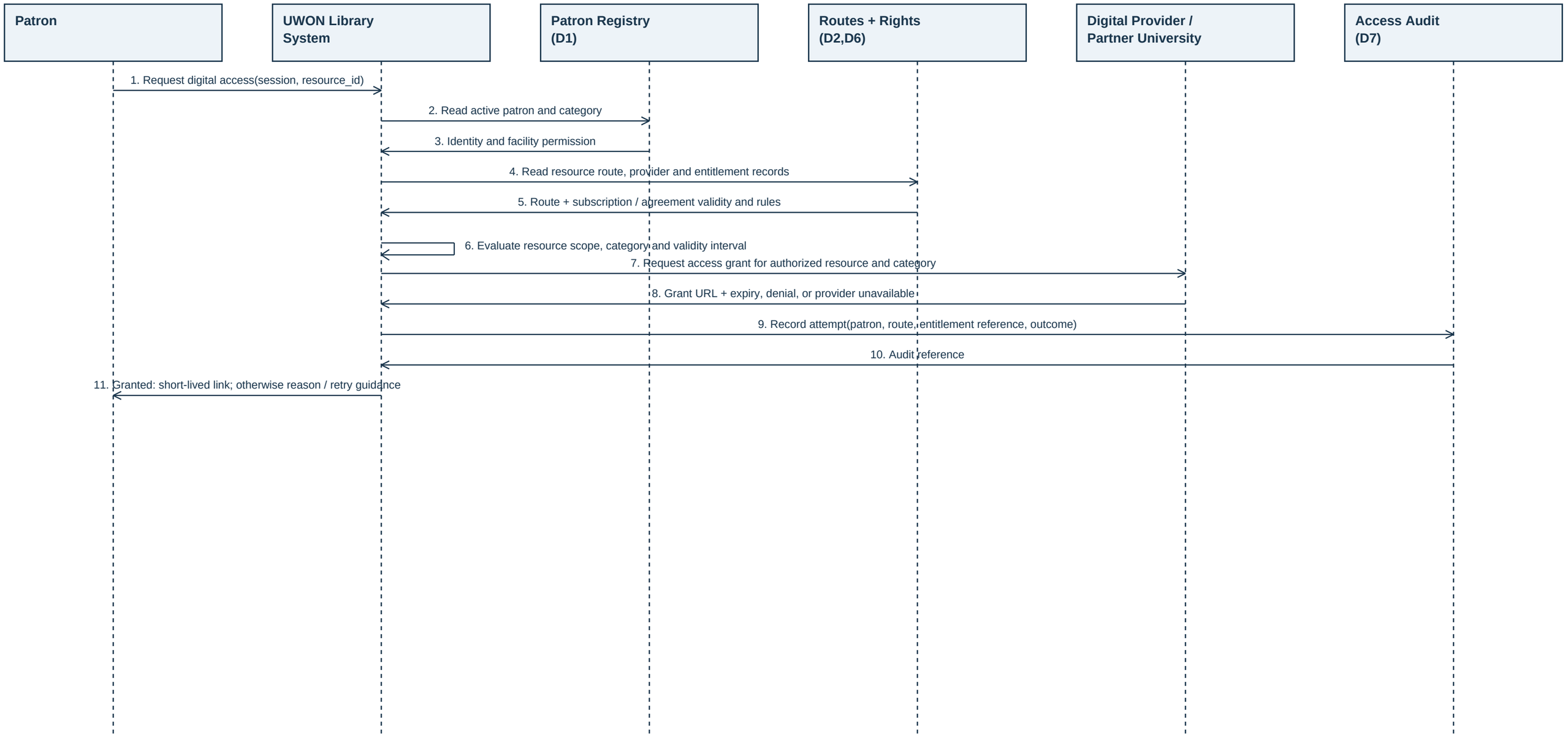
Preconditions: active patron, authorized staff, Ready reservation with allocated copy, and a known loan policy. Loan creation, hold fulfillment and copy status change are atomic; return updates are atomic and closed loans are retained. Alternatives: over limit, wrong patron, missing physical copy or unavailable copy -> refuse before issue; duplicate return -> existing return receipt; damage history supports investigation but does not by itself prove who caused damage.

Time proceeds downward. Logical repositories refer to the same persistent data groups used in the ER and DFD views.

# S3 — Event Trace Diagram

Access a licensed e-journal or partner digital resource

An authenticated patron selects a digital resource and requests access. The system checks the patron category and either an active subscription or an applicable partner-university agreement, then asks the provider for an access grant. A valid provider response produces a short-lived access link and an audit record; denied or unavailable access is recorded without exposing protected content.




### Scenario conditions and alternatives

Precondition: a selected digital catalogue record and active registered patron. Grant requires an active, in-scope Subscription OR Agreement plus provider acceptance; no entitlement means deny before the provider grant request. Each entitlement/provider attempt is retained with Granted, Denied or ProviderUnavailable; identity/facility denials stop before an attempt is created; a granted attempt references exactly one entitlement source, and the audit does not store a reusable access token.

Time proceeds downward. Logical repositories refer to the same persistent data groups used in the ER and DFD views.